

System Conversation Logging

Implementation Plan

1. Purpose

Add a structured, leveled conversation logging system to the LangGraph agent. All agent activity—user prompts, tool calls (Genie, Vector DB), LLM reasoning, and final responses—will be optionally written to a structured logging table in Databricks. The system is designed to be:

- **Non-blocking** — All writes are async fire-and-forget; never slows the agent
- **Toggleable** — A single env var controls everything (AGENT_LOG_LEVEL)
- **Leveled** — Three modes with increasing detail
- **Zero-dependency** — Uses the WorkspaceClient already present in agent.py

2. The Three Log Levels

Level	When to Use	Detail Level
off	Production (Default)	Nothing is written. Zero overhead, zero latency impact.
min	Business Audit	2 rows per turn: user question + final answer with end-to-end duration.
verbose	Engineering Debug	Every activity row — full chain of thought with per-step timing and raw payloads.

Activity Coverage by Level

Activity	off	min	verbose
User Request	NO	YES	YES
LLM Thinking	NO	NO	YES
Call Genie	NO	NO	YES
Call Vector DB	NO	NO	YES
Call Copilot	NO	NO	YES
Final Response	NO	YES	YES

3. Log Table Schema

Table Name: `slf_srvc.test_db.agent_debug_log` (configurable via `DEBUG_LOG_TABLE`)

Column	Type	Nullable	Description
logged_at	TIMESTAMP	NOT NULL	UTC timestamp when the row was written

Column	Type	Nullable	Description
conversation_id	STRING	NOT NULL	Chat session UUID
session_turn	INT	NOT NULL	Increments per user message within a conversation
user_id	STRING	NOT NULL	User email (e.g., javier.delacruz@wheelsup.com)
orchestrator_model	STRING	NOT NULL	LLM endpoint name (e.g., databricks-claude-opus4-7)
activity	STRING	NOT NULL	User Request, LLM Thinking, Call Genie, Call Vector DB, Call Copilot, Final Response
user_prompt	STRING	NULL	The original user question
response	STRING	NULL	Human-readable response text from the activity
duration_ms	INT	NULL	Time taken for this activity in milliseconds
total_tokens	INT	NULL	Total tokens consumed by the LLM for this activity (input + output). Only populated on LLM Thinking and Final Response rows.
status	STRING	NOT NULL	success or error
error_message	STRING	NULL	Error detail if status = error, else null
raw_payload	STRING	NULL	Full JSON payload — only populated at verbose level
retrieved_ids	STRING	NULL	JSON array of business_context.id values matched — e.g. ["1","3","7"]. Only populated on Call Vector DB rows.
similarity_scores	STRING	NULL	JSON array of float scores in same order as retrieved_ids — e.g. [0.92,0.87,0.81]. Only populated on Call Vector DB rows. Low scores (< 0.7) indicate a weak match.

Column Population by Level

Column	min	verbose
logged_at	YES	YES
conversation_id	YES	YES
session_turn	YES	YES
user_id	YES	YES
orchestrator_model	YES	YES
activity	2 values only	all values

5. Example Log Rows

Scenario: WheelsUp member **Juan Smith** asks: *"List of all flighst Dark Vader has taken this year"*

The agent pulls his member profile (Vector DB), queries structured flight records (Genie), formats the result (Copilot), and reasons twice — once to plan, once to synthesize.

User: `juan.smith@wheelsup.com` | conv_id: `conv-7f3a9c` | session_turn: 1

Example A — MIN Level (Business Audit)

Exactly 2 rows are written regardless of how many tool calls fire internally. Row 1 is captured at request arrival; Row 2 captures the final answer and the full end-to-end duration_ms.

logged_at	activity	user_prompt	response	dur. ms	status
2025-05-14 09:12:00.000	User Request	List of all flighst Dark Vader has taken this year	—	—	success
2025-05-14 09:12:04.960	Final Response	List of all flighst Dark Vader has taken this year	Juan Smith has completed 7 flights in 2025 totaling 14.3 hrs across 5 routes.	4,960	success

Example B — VERBOSE Level (Engineering Debug)

All **6 distinct activity types** fire for this query. **LLM Thinking** appears twice (planning + synthesis). Every row carries its own duration_ms and a truncated raw_payload — full JSON in production.

timestamp (UTC)	activity	response / reasoning	dur. ms	status
09:12:00.000	User Request	List of all flighst Dark Vader has taken this year	—	success
09:12:00.215	LLM Thinking	Plan: (1) Fetch Juan Smith's member profile from Vector DB to confirm identity and tier. (2) Query Genie for all 2025 flight records under member ID WU-00482. (3) Call Copilot to format a clean member-facing summary. Then compose final response.	215	success
09:12:00.430	Call Vector DB	{"member_id":"WU-00482","name":"Juan Smith","tier":"Platinum","ytd_hours":14.3,"home_base":"TEB","preferred_aircraft":"G650","joined":"2021-03","status":"active"}	690	success
09:12:01.120	Call Genie	{"total_flights":7,"year":2025,"member":"WU-00482","records":[{"date":"2025-01-08","route":"TEB-MIA","acft":"G650","hrs":2.8}, {"date":"2025-02-14","route":"LAX-ASP","acft":"G550","hrs":3.1}, {"date":"2025-03-01","route":"VNY-DEN","acft":"G650","hrs":1.9}, ...4 more records, total_hrs:14.3]}	1,380	success
09:12:02.500	Call Copilot	{"formatted":"Juan Smith (Platinum / TEB) – 7 flights Jan-May 2025 14.3 hrs total Routes: TEB-MIA, LAX-ASP, VNY-DEN (+4 more) Last flight: May 2 2025 (ORD-TEB, G550) Avg flight duration: 2.04 hrs"}	1,990	success
09:12:04.490	LLM Thinking	All three tools returned successfully. Merging Copilot-formatted summary with Genie flight detail. No missing records detected. Drafting final member-facing response.	175	success
09:12:04.960	Final Response	Juan Smith (Platinum) has completed 7 flights in 2025 totaling 14.3 hrs. Routes include TEB-MIA, LAX-ASP, VNY-DEN and 4 more. Most recent: May 2, 2025 — ORD to TEB on a G550.	4,960	success

Activity Key for This Query

Activity	Role in this conversation
User Request	Captures Juan Smith's raw question the moment it arrives at the agent.
LLM Thinking #1	Plans the tool sequence: Vector DB first for member context, then Genie for flight records, then Copilot to format.
Call Vector DB	Retrieves Juan Smith's member record: tier (Platinum), home base (TEB), preferred aircraft, and YTD stats.
Call Genie	Executes a structured query against the flight records table, filtered to year = 2025 for member WU-00482.
Call Copilot	Transforms raw Genie flight data into a clean, member-facing narrative with route highlights and averages.
LLM Thinking #2	Synthesizes all tool outputs and drafts the final answer — no tool calls needed at this stage.
Final Response	Delivers the completed answer. duration_ms = 4,960 reflects the full wall-clock time from request to response.

Note: raw_payload values above are truncated for readability. In production each row stores the complete, unmodified JSON from the tool. This is the primary artifact for debugging slow calls, unexpected outputs, or tracing errors through the full execution chain.